

Neiv Gupta

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EDUCATION

University of California, Los Angeles

Los Angeles, CA

Bachelor of Science, Computer Science

Expected June 2028

- **Relevant Coursework:** Algorithms & Complexity, Computer Systems Architecture, Software Construction, Electrical & Electronic Circuits, Operating Systems, Logic Design of Digital Systems, Discrete Structures

EXPERIENCE

Software Engineering Intern

June 2026 – Present

Pairwise Technologies (SaaS Startup)

Los Angeles, CA

- Engineered MCP server with FastMCP, exposing DRF views & serializers as OAuth-secured tools for AI agents.
- Built Qdrant vector database dedup pipeline scoring cosine & trigram similarity, with MMR diversity selection.
- Analyzed usage across 100+ backend & frontend services via Django ORM queries to guide engineering decisions.

Software Engineering Intern

Apr. 2025 – Sep. 2025

ThinkScan Technologies (AI Startup)

Pleasanton, CA

- Developed an object detection and scene reasoning AI Agent for threat detection in defense field deployments.
- Built text-to-image retrieval over a FAISS ANN index of CLIP embeddings for efficient imagery database search.
- Enhanced noisy field imagery via Laplacian sharpening and edge-preserving bilateral denoising pre-inference.

RESEARCH

Undergraduate Student Researcher

May 2026 – Present

UCLA Computational Machine Learning Lab, Prof. Cho-Jui Hsieh

Los Angeles, CA

- Researching current audio-LLM defense system robustness against harmful queries and adversarial attacks.
- Engineering a PGD-optimized adversarial audio perturbation that bypasses Qwen2-Audio safety refusals.
- Extending gradient-direction analysis of attack method into the foundation for a new attack-aware defense.

Geographic Information Systems Research Intern

Apr. 2023 - Aug. 2024

Northern Change Research Laboratory, Brown University

Providence, RI

- Executed ResNet-50 transfer learning with TensorFlow/PyTorch, processing Sentinel-2 multi-spectral imagery.
- Engineered data pipelines using GDAL/Rasterio, creating 15K+ labeled training samples via QGIS digitization.
- Contributed to glacial ice loss and sea level rise projections achieved via geodetic mass balance calculations.

Computer Vision Student Researcher

Feb. 2024 - June 2024

Argonne National Laboratory

Lemont, IL

- Deployed TensorFlow/PyTorch CNN pipelines for wildfire response and drought monitoring.
- Built CLIP-based zero-shot classification model achieving 93% mAP for wildfire and drought imagery datasets.
- Fine-tuned OpenCLIP ViT-B/32 multi-modal models on imagery datasets for improved accuracy.

PROJECTS

CudaFire 🔥 | *CUDA, C++17, C, CMake, GDAL, OpenGL*

Dec. 2025 – Jan. 2026

- Built GPU-accelerated wildfire spread simulator using CUDA and the Rothermel fire behavior model.
- Processed 8.7 million terrain cells in parallel using 8-connected cellular automaton on RTX 3080.
- Optimized CUDA kernels with 16×16 thread blocks, achieving 7,643× real-time simulation performance.

BruinMarket 🌐🔒 | *Go, React, PostgreSQL, WebSockets, Docker, Railway, Vercel*

Oct. 2025 – Nov. 2025

- Launched full-stack UCLA-exclusive student marketplace with real-time peer-to-peer transactions and messaging.
- Architected backend using Go with PostgreSQL database and JWT authentication and email verification.
- Deployed production app on Railway and Vercel with custom domain configuration and CI/CD pipeline.

YUM 🍔 | *React Native (Expo), Node.js, Express.js, MongoDB Atlas, REST APIs*

Mar. 2025 – Jun. 2025

- Developed UCLA mobile dining app providing live dining updates, commenting workflows, and personal profiles.
- Architected full-stack MERN mobile app with JWT-based authentication and real-time state synchronization.
- Implemented RESTful API endpoints with Express.js middleware and MongoDB aggregation pipelines.

TECHNICAL SKILLS

Languages: Java, Python, C, C++, Go, JavaScript, TypeScript, HTML/CSS, XML, JSON, Swift, Bash, SQL, CUDA

Frameworks: PyTorch, TensorFlow, LangChain, React, React Native, Node.js, Express.js, Gin, Django, Celery

Tools & Concepts: Git, GitHub, Docker, Kubernetes, MongoDB, PostgreSQL, Railway, Vercel, FastMCP, CI/CD

Libraries: NumPy, pandas, Matplotlib, Scikit-learn, OpenCV, CLIP, OpenCLIP, YOLO, XGBoost, Tailwind CSS